

นี่คือสคริปต์การนำเสนอที่ปรับแก้ใหม่ครับ โดยในส่วนของ Slide 1-7 ผมได้ทำการ **ย่อเนื้อหาให้กระชับขึ้น**

(Condensed) เน้นเฉพาะ Key Points ที่สำคัญที่สุด เพื่อให้มีเวลาไปอธิบายรายละเอียดใน Slide 8-10 อย่างเต็มที่ครับ

ระดับภาษาที่ใช้คือ A2 ถึง B1 ซึ่งจะเน้นประโยคที่สั้น เข้าใจง่าย ตรงไปตรงมา ทำให้ทั้งอาจารย์และเพื่อนๆ ชาวไต้หวันสามารถติดตามเนื้อหาได้ทันแน่นอนครับ

Part 1: The Core Proposal (Condensed)

Slide 1: Title Slide "Good morning, Professor and everyone. My name is Lin Zhilong. Today, I will present my thesis methodology: 'Robust SOH Estimation Framework using AE-BPNN and Transfer Learning'."

Slide 2: Outline (คลิกเปลี่ยนสไลด์) "Here is my outline. I will quickly cover the research gaps, my selected AE-BPNN model, three proposed scenarios, my data strategy, and the project roadmap."

Slide 3: Identifying the Gaps (คลิกเปลี่ยนสไลด์) "Let's look at the gaps in previous work. First, information loss from compressing data into an 8x8 matrix. Second, data scarcity because we only have 8 cells in our lab, which causes overfitting. Third, the data is heavily biased towards 25 degrees Celsius, limiting real-world application."

Slide 4: Model Selection (คลิกเปลี่ยนสไลด์) "To address the gaps, I selected the AE-BPNN as my core architecture.

First, the Autoencoder reduces the data size into 20 key features.

Second, **why do I choose BPNN?** There are two main reasons.

1. Battery aging is highly non-linear. BPNN is a great tool for learning complex and non-linear relationships.
2. Compared to deep networks, a simple BPNN is very lightweight. It requires low computational cost but still provides a high R-squared score.

Finally, I will add a robustness test by injecting noise to simulate real-world sensor noise."

Slide 5: Proposed Experiments (คลิกเปลี่ยนสไลด์) "Next, I propose three experimental scenarios. **Why do we need 3 scenarios?** Because they are designed step-by-step to cover all the real-world problems.

- **Scenario A** is the Baseline. It proves that our model is accurate under normal conditions.
- **Scenario B** is the Robustness test. It proves the model is strong and stable even when the sensor data is noisy.
- **Scenario C** is the Data Scarcity test. It proves we can overcome the lack of lab data by using Transfer Learning.

Together, these 3 scenarios cover accuracy, stability, and real-world limits. This makes the SOH estimation framework complete and reliable."

Slide 6: Open-Source Data Strategy (คลิกเปลี่ยนสไลด์) "For the open-source data, I use the NASA PCoE dataset. I will pre-train the model with this large dataset, then fine-tune it with our 8 lab cells. This lets me start coding immediately."

Slide 7: Tentative Project Roadmap (คลิกเปลี่ยนสไลด์) "My roadmap has 4 phases. Phase 1 is data prep. Phase 2 covers Scenarios A and B. Phase 3 covers Scenario C. Finally, Phase 4 is data analysis and thesis writing."

Part 2: Initial Progress & Next Steps (Detailed)

Slide 8: Progress: Data Extraction & Consolidation (คลิกเปลี่ยนสไลด์) "Now, let's look at my current progress. For Phase 1, I used MATLAB to extract data from 4 NASA battery cells: B0005, B0006, B0007, and B0018.

I extracted these 4 cells together because they were all tested at the same room temperature. Combining them gives the model more diverse patterns to learn from.

As you can see, I only extracted 4 columns: Battery ID, Cycle, Re_Z, and Negative Im_Z. Why only these 4? Because our model only needs the real and imaginary impedance values to estimate SOH. Other data formats like voltage or current are not needed for this specific model. This keeps the dataset clean and simple for the AI."

Slide 9: Proof of Concept: Architecture Verification (คลิกเปลี่ยนสไลด์) "Next, this is my PyTorch code for the Autoencoder blueprint. As you can see in the terminal, it runs successfully with over 34,000 records.

Let me explain the design. First, why do I use **Linear layers**? Because my input is a flat 1D array of 256 features, not a 2D picture. Linear layers are lightweight and perfect for this data format.

Second, why do I use **ReLU**? Battery aging is very complex. ReLU helps the model learn these complex, non-linear patterns. Without it, the model would be too simple and cannot predict SOH accurately.

Finally, you can see the data size drops from 256 to 128, then 64, and finally to 20 latent features. 20 is our target bottleneck. This step-by-step reduction forces the AI to keep only the most important battery health information and throw away the noise."

Slide 10: Immediate Next Steps (คลิกเปลี่ยนสไลด์) "For my immediate next steps I will do three things.

Step 1: Data Interpolation. I will write code to transform the 1D impedance points into the 16x16 matrix format for the model input. Step 2: Train the Autoencoder. I will feed the NASA dataset into the model to extract the 20 latent features. Step 3: Connect to BPNN. I will connect this trained encoder to the BPNN to start predicting the actual SOH.

Thank you for listening.